# ASSIGNMENT (NOV 3)

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## 1. Local cohomology of injective modules (reading material)

Let $A$ be a Noetherian ring, let $I \subseteq A$ be an ideal, and let $E$ be an injective $A$-module. We claim that $R\Gamma_I(E)$ is concentrated in degree zero[1] and that $\mathrm{H}^0_I(E)$ is injective. The latter assertion follows from [Stacks, Tag 08XW], so it remains to establish the concentration in degree zero. Here are four ways to see this. I asked chatGPT to write the first three and wrote the last one myself.

(1) *Sheaf-theoretic argument.* Put $X = \operatorname{Spec} A$, $Z = V(I)$, and $U = X \setminus Z$. The sheaf $\widetilde{E}$ is flasque [Har77, Chapter III, Proposition 3.4]. so $\mathrm{H}^i(U, \widetilde{E}) = 0$ for $i > 0$, and the restriction map
$$E = \Gamma(X, \widetilde{E}) \longrightarrow \Gamma(U, \widetilde{E})$$
is surjective. The local cohomology triangle
$$R\Gamma_I(E) \longrightarrow E \longrightarrow R\Gamma(U, \widetilde{E}) \longrightarrow R\Gamma_I(E)[1]$$
therefore shows that $\mathrm{H}^i_I(E) = 0$ for all $i > 0$.

(2) *Classification of injective modules.* Every injective $A$-module is a direct sum of indecomposable injective modules $E_A(A/\mathfrak{p})$ (injective hull of $A/\mathfrak{p}$), with $\mathfrak{p} \in \operatorname{Spec} A$. Since $R\Gamma_I$ commutes with direct sums, it suffices to consider $E = E_A(A/\mathfrak{p})$.

If $I \subseteq \mathfrak{p}$, then $E$ is $I$-power torsion. Writing $I = (f_1, \ldots, f_r)$, we have $E_{f_i} = 0$ for every $i$, so the augmented Čech complex computing $R\Gamma_I(E)$ reduces to $E$ in degree zero. If $I \not\subseteq \mathfrak{p}$, choose $f \in I \setminus \mathfrak{p}$. Multiplication by $f$ on $E$ is invertible, and hence $R\Gamma_I(E) = 0$. Thus, in either case, the desired concentration follows.

(3) *Elementary splitting argument.* Set $T = \mathrm{H}^0_I(E)$. Since $T$ is injective, its inclusion into $E$ splits, giving
$$E = T \oplus E', \qquad \mathrm{H}^0_I(E') = 0.$$
The augmented Čech complex shows that $R\Gamma_I(T) \simeq T$, since $T$ is $I$-power torsion. It therefore suffices to prove $R\Gamma_I(E') = 0$.

Write $I = (f_1, \ldots, f_r)$. The natural map
$$E' \longrightarrow \bigoplus_{i=1}^{r} E'_{f_i}$$
is injective: its kernel is precisely $\mathrm{H}^0_I(E')$. Since $E'$ is injective, this inclusion splits. On each $E'_{f_i}$, an element of $I$ acts invertibly, so $R\Gamma_I(E'_{f_i}) = 0$. Applying $R\Gamma_I$ to the splitting shows that $R\Gamma_I(E')$ is a direct summand of zero, and hence vanishes.

(4) *The derived/canonical version of (3).* Let $T$ be as above, and let $C = \operatorname{cofib}(R\Gamma_I(E) \to E)$, so we have a distinguished triangle
$$R\Gamma_I(E) \longrightarrow E \longrightarrow C \longrightarrow +1,$$

[1] Hence $R\Gamma_I(-)$ really is the right derived functor of $\mathrm{H}^0_I(-)$. One can also show that for every free $A$-module $F$ the completion $\widehat{F}^I$ is concentrated in degree 0, so $\widehat{(-)}^I$ is the left derived functor of $\mathrm{H}^0\left(\widehat{(-)}^I\right)$, which one can show is identified with the classical completion. We never needed these facts.

which induces a distinguished triangle

$$\tau^{\geq 1}R\Gamma_I(E) \longrightarrow E/T \longrightarrow C \longrightarrow +1.$$

As seen above $E/T$ is injective, so $\mathrm{Hom}_{\mathrm{D}(A)}(\tau^{\geq 1}R\Gamma_I(E), E/T) = 0$, whence in $\mathrm{D}(A)$ we have

$$C \simeq (E/T) \oplus \tau^{\geq 1}R\Gamma_I(E)[1].$$

As $R\Gamma_I$ is idempotent we have $R\Gamma_I C = 0$. Also $\tau^{\geq 1}R\Gamma_I(E)[1] \in \mathrm{D}_{I^\infty\text{-tors}}(A)$. Hence $\tau^{\geq 1}R\Gamma_I(E) = 0$.

## 2. Injective and projective dimension

Made from handwritten notes by myself with chatGPT and editing.

**Remark 2.1.** It is not difficult to prove the following by transfinite induction (Baer's criterion). Let $A$ be a ring and $E \in \mathrm{D}^{\heartsuit}(A)$. Then $E$ is injective if and only if

$$\mathrm{Ext}^1_A(A/\mathfrak{a}, E) = 0 \qquad \text{for every ideal } \mathfrak{a} \subseteq A.$$

Another transfinite induction argument tells us that every $M \in \mathrm{D}^{\heartsuit}(A)$ is contained in some injective $E \in \mathrm{D}^{\heartsuit}(A)$.

For a complex $X \in \mathrm{D}(A)$ and $\alpha, \beta \in \mathbf{Z}$, it is not difficult to show that the following are equivalent:

(1) For every finitely generated $M \in \mathrm{D}^{\heartsuit}(A)$, we have
$$\mathrm{RHom}_A(M, X) \in \mathrm{D}^{[\alpha,\beta]}(A).$$
(2) $X$ is represented by a complex of injective $A$-modules in degrees $[\alpha, \beta]$.

In this case, we say that $X$ has *injective amplitude* in $[\alpha, \beta]$. Taking $M = A$ we see in particular $X \in \mathrm{D}^{[\alpha,\beta]}(A)$.

What do we know about injective amplitude? Note that it is trivial that

$$\text{For all } X \in \mathrm{D}^{\geq\alpha}(A) \text{ and all } M \in \mathrm{D}^{\heartsuit}(A), \mathrm{RHom}_A(M, X) \in \mathrm{D}^{\geq\alpha}(A). \tag{2.1}$$

**Problem 1.** Let $(A, \mathfrak{m}, k)$ be a Noetherian local ring.

(1) Let $X \in \mathrm{D}_{\mathrm{comp}}(A)$ or $\mathrm{D}^+_{\mathrm{coh}}(A)$, and let $\beta \in \mathbf{Z}$. Show that if
$$\mathrm{RHom}_A(k, X) \in \mathrm{D}^{\leq\beta}(A),$$
then for every $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$, we have
$$\mathrm{RHom}_A(M, X) \in \mathrm{D}^{\leq\beta-\operatorname{depth} M}(A).$$
(2) Let $X \in \mathrm{D}_{\mathrm{comp}}(A)$ or $\mathrm{D}^+_{\mathrm{coh}}(A)$, and let $\alpha \in \mathbf{Z}$. Show that if
$$\mathrm{RHom}_A(k, X) \in \mathrm{D}^{\geq\alpha}(A),$$
then for every $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$, we have
$$\mathrm{RHom}_A(M, X) \in \mathrm{D}^{\geq\alpha-\dim \operatorname{Supp} M}(A).$$

Consequently, if $X \in \mathrm{D}_{\mathrm{comp}}(A)$ or $\mathrm{D}^+_{\mathrm{coh}}(A)$ is such that $\mathrm{RHom}_A(k, X) \in \mathrm{D}^{[\alpha,\beta]}(A)$, then $X \in \mathrm{D}^{\mathrm{b}}(A)$, and $X$ has injective amplitude in $[\max\{\alpha - \dim A, c\}, \beta]$, for any $c \in \mathbf{Z}$ such that $X \in \mathrm{D}^{\geq c}(A)$.

**Problem 2.** Let $(A, \mathfrak{m}, k)$ be a Noetherian local ring.

(1) Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$, and let $\beta \in \mathbf{Z}$. Show that if
$$\mathrm{RHom}_A(k, X) \in \mathrm{D}^{\leq\beta}(A),$$
then for every $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$, we have
$$\mathrm{RHom}_A(M, X) \in \mathrm{D}^{\leq\beta}(A).$$

(2) Let $X \in \mathrm{D}^+_{\mathrm{tors}}(A)$, and let $\alpha \in \mathbf{Z}$. Then, we have seen in the Oct 20 meeting (Lemma 5.1 in the notes) that if

$$\mathrm{RHom}_A(k, X) \in \mathrm{D}^{\geq \alpha}(A),$$

then $X \in \mathrm{D}^{\geq \alpha}(A)$, hence for every $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$, we have

$$\mathrm{RHom}_A(M, X) \in \mathrm{D}^{\geq \alpha}(A) \text{ (cf. (2.1))}.$$

For $M \in \mathrm{D}^{\heartsuit}(A)$, let

$$\mathrm{pd}_A M = \min\{j \mid \text{for all } i > j,\ \mathrm{Ext}^i_A(M, -) = 0\}, \tag{2.2}$$

$$\mathrm{id}_A M = \max\{j \mid \text{for all } i > j,\ \mathrm{Ext}^i_A(-, M) = 0\}. \tag{2.3}$$

Let $(A, \mathfrak{m}, k)$ be Noetherian local. In the October 27 meeting, we saw, using minimal resolutions, that if $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$, then

$$\mathrm{pd}_A M = \max\{j \mid \mathrm{Tor}^A_j(k, M) \neq 0\} = \min\{j \mid \mathrm{Tor}^A_j(k, M) = 0\} - 1. \tag{2.4}$$

From the previous problem we see that if $M \in \mathrm{D}^{\heartsuit}_{\square}(A)$, where $\square \in \{\mathrm{tors}, \mathrm{comp}, \mathrm{coh}\}$, then

$$\mathrm{id}_A M = \max\{j \mid \mathrm{Ext}^j_A(k, M) \neq 0\}. \tag{2.5}$$

**Problem 3.** Let $f \in \mathfrak{m}$ be a nonzerodivisor in $A$. Let $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A)$ be such that $fM = 0$. Show that

$$\mathrm{pd}_{A/fA} M + 1 = \mathrm{pd}_A M \qquad \text{if } \mathrm{pd}_{A/fA} M < \infty, \tag{2.6}$$

$$\mathrm{id}_{A/fA} M + 1 = \mathrm{id}_A M \qquad \text{if } \mathrm{id}_{A/fA} M < \infty. \tag{2.7}$$

**Problem 4.** Let $0 \neq M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(A) \cap \mathrm{D}_{\mathrm{tors}}(A)$, i.e. let $M$ be a nonzero finite-length $A$-module.

(1) If $\mathrm{pd}_A M < \infty$, show that $A$ is Cohen–Macaulay and $\mathrm{pd}_A M = \dim A$.
(2) If $\mathrm{id}_A M < \infty$, show that $A$ is Cohen–Macaulay and $\mathrm{id}_A M = \dim A$.
(3) Conversely, assume that $A$ is Cohen–Macaulay. Show that such an $M$ exists. You may take different modules $M$ for projective dimension and injective dimension, respectively.

Note: we unfortunately cannot use Problem 3 for induction, as we do not know if $\mathrm{pd}_{A/fA} M < \infty$. Therefore, (1) is likely too difficult for us (cf. [BH98, §9.4]). However, assuming (1) for granted, (2) should be within reach. Hint: consider $\mathrm{Hom}_A(M, E)$.

One may hope to check projectivity of general modules similarly, using $\mathrm{Ext}^i_A(-, k)$ or $\mathrm{Tor}^A_i(-, k)$. This is impossible. Indeed, if $A$ is a discrete valuation ring, then all flat $A$-modules $F$ satisfy $\mathrm{Ext}^i_A(F, k) = 0 = \mathrm{Tor}^A_i(F, k)$ for all $i > 0$. To see this, consider a uniformizer $\pi$, which gives an exact sequence

$$0 \longrightarrow k \longrightarrow E \xrightarrow{\times \pi} E \longrightarrow 0,$$

hence $\mathrm{RHom}_A(F, k)$ is the fiber of the $\times\pi$ map on $\mathrm{RHom}_A(F, E) = \mathrm{Hom}_A(F, E)$. But as $F$ is flat, tensor-hom adjunction tells $\mathrm{Hom}_A(F, E)$ is injective, hence divisible, so $\mathrm{RHom}_A(F, k) \in \mathrm{D}^{\heartsuit}(A)$.

One could ask: if a $\mathbf{Z}$-module $F$ satisfies $\mathrm{Ext}^1_A(F, \mathbf{Z}) = 0$[2], is $F$ projective? This is known as the *Whitehead Problem* and is independent of ZFC, the first purely algebraic problem of such nature.

**Problem 5.** Let $(A, \mathfrak{m}, k)$ be an Artinian local ring (*i.e.* Noetherian and $\mathfrak{m}^N = 0$ for some $N$).

(1) Show that $\mathrm{D}_{\mathrm{comp}}(A) = \mathrm{D}(A) = \mathrm{D}_{\mathrm{tors}}(A)$.
(2) Show that $E$, the injective hull of $k$, is in $\mathrm{D}_{\mathrm{coh}}(A)$.
(3) Show that if $M \in \mathrm{D}^{\heartsuit}(A)$ satisfies $\mathrm{Ext}^1_A(M, k) = 0$, then $M$ is flat.
(4) Show that every flat $A$-module is projective.

In particular, you can test projectivity with $\mathrm{Ext}^1_A(-, k)$.

[2]which implies $F$ is flat and $\mathrm{Ext}^i_A(F, M) = 0$ for all $i > 0$ and all $M \in \mathrm{D}^{\heartsuit}_{\mathrm{coh}}(\mathbf{Z})$

## References

[BH98] W. Bruns and H.J. Herzog. *Cohen–Macaulay Rings, 2nd edition*. Cambridge University Press, 1998.

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. Springer New York, NY, 1977.

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